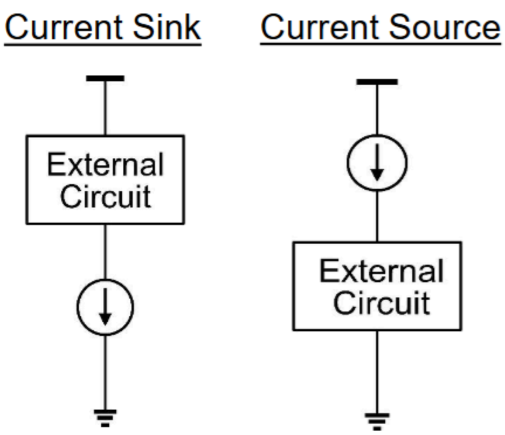
Introduction

Why Current Source?

- We have briefly introduced the simple current source/mirror and cascode current source.
- We have seen in previous lectures that current course and mirror are used as “active load” in various amplifiers to increase the gain and relax the trade off with output swing in resistive load amplifiers.
- In later lectures, we’ll see that the current source and mirror can be used to bias amplifiers.
- In this lecture, analysis and design of the current source and mirrors will be discussed.

- if more accurate current source is needed, active current source is needed

Ideal Current Source / Sink



- Whether it is a current source or sink depends on whether the current flows in or out of the external circuit
- Current source → Current flows into the external circuit
- Current sink → Current flows out of the external circuit
- We often call both source and sink a "Current Source".

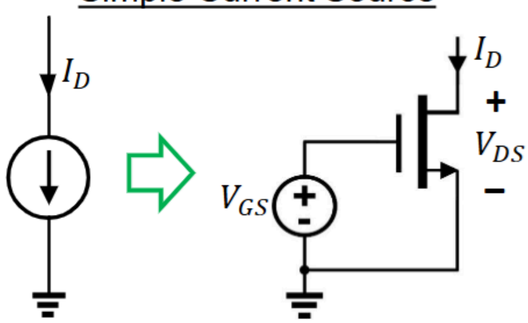
Ideal current source / sink

- An Ideal current source (sink) provide a constant current that is NOT dependent on the voltage across itself
- An ideal current source (sink) has infinite output resistance

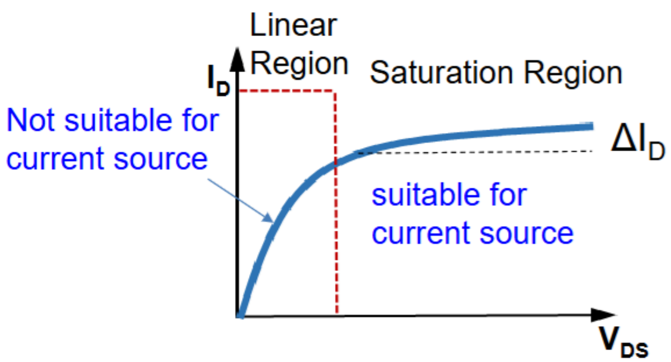
- for ideal current source there is no difference between sinking or sourcing
- but for mosfet that is not possible, and it's based on the Nfet or Pfet
- if the current source is closer to ground then its own voltage headroom required to stay in saturation region will be reference to ground, vice versa
- For AC analysis, current source is opened, so its ideally infinite output resistance

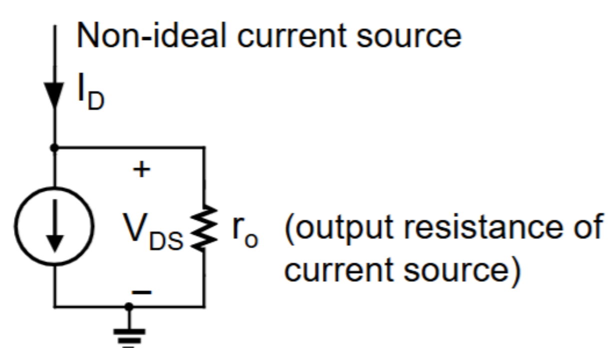
Realization of Current Source

Simple Current Source



- MOSFET in saturation region with a fixed V_{GS} can be used to implement a current source (I_D)
- I_D is a function of V_{GS}
- Due to the channel length modulation effect, this current source has finite output resistance.
- A minimum output voltage, that is, $V_{DS} = V_{OD}$ is required to keep the MOSFET in saturation.





- MOSFET must be in saturation because that's when the current solely depends on the the gate voltage, not the VDS (ignoring channel modulation effect)

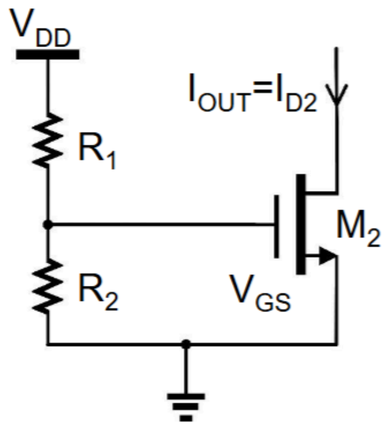
- Linear Region $v_{GS} \geq V_{TH}$ & $v_{DS} \leq v_{DSsat} = v_{GS} - V_{TH}$ [$v_{GD} \geq V_{TH}$]
- $i_D = \mu_n \frac{W}{L} C_{ox}[(v_{GS} - V_{TH})v_{DS} - \frac{1}{2}v_{DS}^2]$
- Saturation Region $v_{GS} \geq V_{TH}$ & $v_{DS} \geq v_{DSsat} = v_{GS} - V_{TH}$ [$v_{GD} \leq V_{TH}$]
- $i_D = i_{Dsat} = \frac{1}{2}\mu_n \frac{W}{L} C_{ox}(v_{GS} - V_{TH})^2$
- Saturation Region with channel length modulation
- $i_D = K_n(v_{GS} - V_{TH})^2(1 + \lambda v_{DS})$

- Due to channel modulation effect there is output resistance
  - increase L to reduce this effect, then the current source acts more ideal
- And there is a limit to the minimum voltage drop across the MOSFET, otherwise it will go into linear region and act like a resistor
  - imagine a resistive load on top of the current source shown
  - to sink more current the voltage drop must be lower
  - and this will cause it to go into linear region
- So realization of current source with MOSFET has its limitation

Current Source Implementation

- A simple current source circuit is shown below. The MOS transistor is biased with a resistive voltage divider (R_1, R_2).
- This method has problems as it is sensitive to V_{DD}, R_1, R_2 and V_{TH}. All of these parameters change with PVT (Process, Voltage, Temperature) variation.

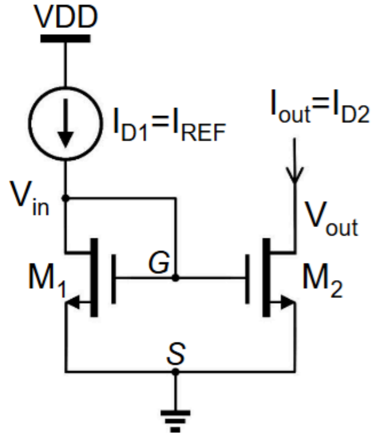
$$I_{D2} = \frac{1}{2}\left(\frac{W}{L}\right)_2 \mu C_{ox}(V_{GS} - V_{TH})^2 (1 + \lambda V_{DS2})$$
$$= \frac{1}{2}\left(\frac{W}{L}\right)_2 \mu C_{ox}\left(\frac{V_{DD} \cdot R_2}{R_1 + R_2} - V_{TH}\right)^2 (1 + \lambda V_{DS2})$$



- A good analog design should be insensitive to PVT variation.
- A simple current mirror can overcome these problems.

- this is very sensitive to PVT
- absolute resistor fabrication is good for 10 to 20%, and the ratio is better, but it is still not good enough
- and Vdd is self can change also, unless power supply is very stable
- temperature dependence is also very important, because it depends on the application, and how big a range of temperatures the final design can handle
- So instead of using resistors, we use current mirror, where one good current source can be mirrored to many

Current Mirror



$$I_{REF} = I_{D1} = \frac{1}{2}k_n'\left(\frac{W}{L}\right)_1 (V_{GS} - V_{TH})^2(1 + \lambda V_{DS1})$$

$$I_{out} = I_{D2} = \frac{1}{2}k_n'\left(\frac{W}{L}\right)_2 (V_{GS} - V_{TH})^2(1 + \lambda V_{out})$$

$$\frac{I_{out}}{I_{REF}} = \frac{I_{D2}}{I_{D1}} = \frac{\left(\frac{W}{L}\right)_2 (1 + \lambda V_{out})}{\left(\frac{W}{L}\right)_1 (1 + \lambda V_{GS1})}$$

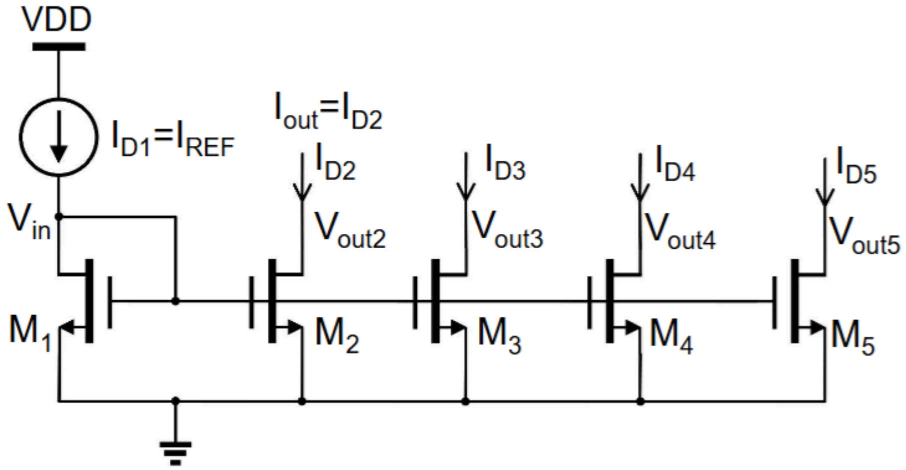
- A current mirror has at least two branches. A constant reference current I_{REF} flows through one branch and the current in the other branch "mirrors" the reference current
- The key idea of the current mirror is to set V_{GS} of M_2 to a fixed value by diode connected transistor M_1
- Since both transistors share the same V_{GS}, the output current is

$$I_{out} = \frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1} I_{REF} \qquad (V_{in} = V_{out})$$

- Since both transistors are from the same process and under the same temperature variations, the PVT effect will be cancelled
- This current source utilizes matched devices to cancel PVT effect. This is a common way to deal with PVT in analog circuit design

- using matched devices, to cancel out PVT variation to make the design more robust
  - matched in terms of process variation, powered by the same voltage and on the same die or near each other physically so the temperature is the same
- The derivation above assumes that the Vth is the same because it is the same type of transistor on the same fabrication process
  - but if the transistor sizing is different the threshold can be different also
- The derivation also neglects the channel modulation effect
- Then and only then the current ratio becomes the sizing ratio
  - otherwise if the mosfet is not matched the Vth, kn and λ are all different

Current Mirror with Multiple Output Currents



- More transistors can be added to provide multiple output currents or current sources.
- By designing the transistors with different aspect ratio, different currents can be mirrored out from the same reference current, and

$$I_{Dx} = \frac{\left(\frac{W}{L}\right)_x}{\left(\frac{W}{L}\right)_1} I_{REF} \qquad \text{Current gain: } \alpha_x = \frac{\left(\frac{W}{L}\right)_x}{\left(\frac{W}{L}\right)_1}$$

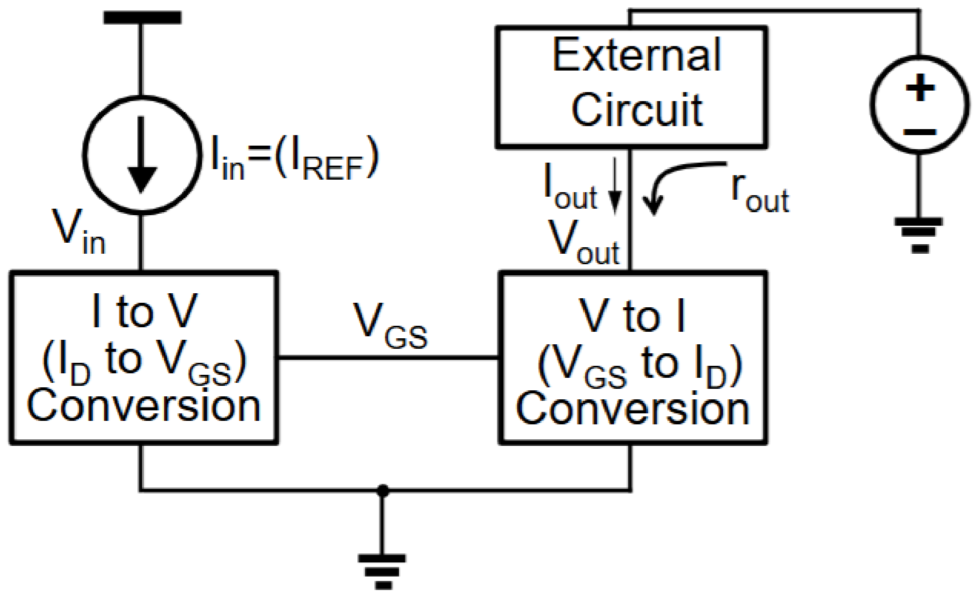
x=2, 3, 4 ,5…

Assume $V_{outx} = V_{GS1}$

- master bias to many current out

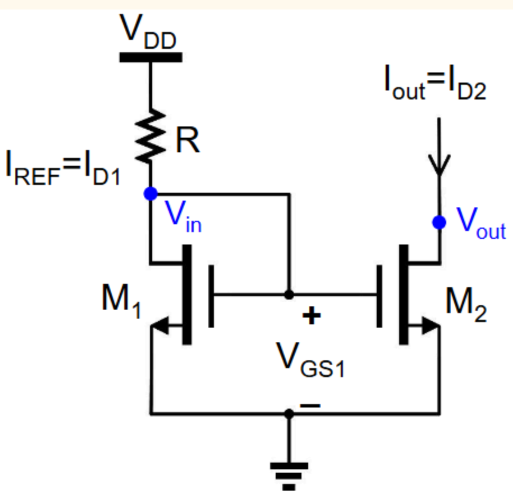
Another Way to Look at Current Mirror

- The basic current mirror can be viewed as a I-to-V-to-I converter.
- Current mirror specifications:
  - Current gain $\alpha = I_{out}/I_{in}$
  - small-signal output resistance r_o
  - Minimum output voltage (for transistor in saturation), V_{out_min}



- from current create a voltage which then create a current

Simple Current Mirror



For diode-connected transistor M_1: $V_{DS1} = V_{GS1} > V_{GS1} - V_{TH1}$

Hence, M_1 is always in saturation region

$$I_{D1} = \frac{1}{2}\mu C_{ox}\frac{W}{L}(V_{GS1} - V_{TH1})^2\,(1 + \lambda V_{DS1}) = \frac{V_{DD} - V_{GS1}}{R}$$

For a given I_{D1}, we can solve the cubic equation to find V_{GS}, and then value of R can be calculated

If the channel length modulation term is ignored, we can calculate the value of R.

$$I_{D1} = \frac{1}{2}\mu C_{ox}\frac{W}{L}(V_{GS1} - V_{TH1})^2 = \frac{V_{DD} - V_{GS1}}{R}$$

What is the condition for M2 to be in saturation?
$V_{DS2} = V_{out} > V_{GS1} - V_{TH1} \rightarrow$ minimum output $V_{out\_min} \geq V_{GS1} - V_{TH2} = V_{OD2}$

- this is the same as EE2027

With accurate model that includes channel length modulation, we have:

$$I_{D1} = \frac{1}{2}\mu C_{ox}\left(\frac{W}{L}\right)_1 (V_{GS1} - V_{TH1})^2(1 + \lambda_1 V_{GS1})$$

$$I_{D2} = \frac{1}{2}\mu C_{ox}\left(\frac{W}{L}\right)_2 (V_{GS1} - V_{TH2})^2(1 + \lambda_2 V_{GS2})$$

Since both transistors are fabricated with the same process, they have the same parameter $k' = \mu C_{ox}$ and V_{TH}, hence

$$\frac{I_{out}}{I_{REF}} = \frac{I_{D2}}{I_{D1}} = \frac{\left(\frac{W}{L}\right)_2 (1 + \lambda_2 V_{out})}{\left(\frac{W}{L}\right)_1 (1 + \lambda_1 V_{in})}$$

It is desirable to choose the same channel length for M_1 and M_2, so that $\lambda_1 = \lambda_2$

$$\frac{I_{out}}{I_{in}} = \frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1}$$

(Only if V_{out}=V_{in})

- NOTE: with different sizing, VTh is actually different so this slide is simplified
- BUT, this slides include the channel modulation effect
- and the current ratio at the end is only true iff Vout = Vin, (denoted in the diagram in blue)
    - this is not possible in actually application because Vouit depends on the external circuit, and for amplifier it is always changing and the ratio is no longer true
    - cannot cancel out the λ

Current Mirror Analysis and Errors

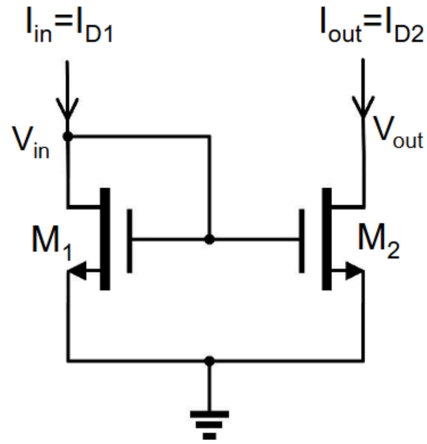
- Desirable specifications:
    - High output resistance r_o → I_{out} is less dependent on V_{out}
    - Minimum output voltage V_{out_min} → Large output swing
    - Accurate current gain → Well defined output current
- The first two parameters depend on the chosen current source structure and the design parameters
- The inaccuracy of the current gain may be caused by two types of errors:
    - systematic error → the error caused by circuit structure and improper design
    - Random error → the error due to process variations
- These two types of error are commonly seen in analog integrated circuits. Robust design is to minimize both systematic and random errors

- can see that from the simple current mirror before, the current is actually dependent on a lot of factors, resulting in error in the current

Gain Error (Systematic Error)

Systematic gain error definition: $E_{rr} = \frac{I_{OUT} - I_{OUT,ideal}}{I_{OUT,ideal}}$

Assume $\lambda_1 = \lambda_2$



$$\frac{I_{out}}{I_{in}} = \frac{I_{D2}}{I_{D1}} = \frac{\left(\frac{W}{L}\right)_2 (1+\lambda V_{out})}{\left(\frac{W}{L}\right)_1 (1+\lambda V_{in})} \qquad I_{out} = \frac{\left(\frac{W}{L}\right)_2 (1+\lambda V_{out})}{\left(\frac{W}{L}\right)_1 (1+\lambda V_{in})} I_{in}$$

Gain error:

$$I_{out,ieeal} = \frac{(W/L)_2}{(W/L)_1} I_{in} \quad \text{and assume} \quad \frac{(W/L)_2}{(W/L)_1} = 1$$

$$E_{rr} = \frac{I_{out} - I_{out,ideal}}{I_{out,ideal}} = \frac{(1+\lambda V_{out})}{(1+\lambda V_{in})} - 1 = \frac{\lambda}{1+\lambda V_{in}} (V_{out} - V_{in})$$

If $\lambda V_{in} \ll 1$ $\qquad E_{rr} \approx \lambda (V_{out} - V_{in})$

Systematic gain error can be

– reduced by using long channel devices (small λ), or

– completely eliminated by making $V_{out} = V_{in}$

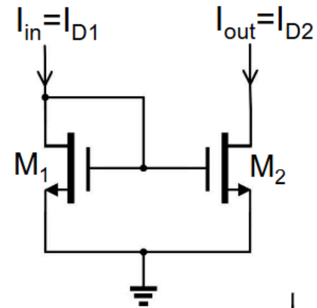
- this is the error due to λ, or due to the difference of the two voltages
- so can be handled
  - reduction in error with big L and small λ
  - elimination of error with Vout = Vin, somehow force them to be the same
    - the somehow is [cascode current mirror]

Mismatch Error (Random Error)

- Matched devices mean all the devices are designed to have matched parameters.
- Mismatch → parameter differences between matched devices that are designed to be matched
- Causes of mismatch
  – Process variation caused geometry mismatch, such as channel length and width, and physical parameter mismatch, such as doping concentration
  – Material non-uniformity
  – Pattern shift (Mask misalignment in photolithography process)
- We'll focus on geometry mismatch caused by process variation

- this is random error in the sizing and whatnot during fabrication
  - for doping concentration variation, it would affect large devices more
- circuit designer cannot reduce process variation, so must design the circuit to be robust against this
  - and its only mainly the geometry mismatch we can handle

- A current mirror with current ratio of two is designed. Assume $V_{IN}=V_{OUT}$ to eliminate the channel length modulation effect.
- Due to mismatch between M_1 and M_2, the output current may not be exactly two times of the input current. See examples below.
- 3 ways to implement:
  (assumes same ΔW and ΔL for all transistors)

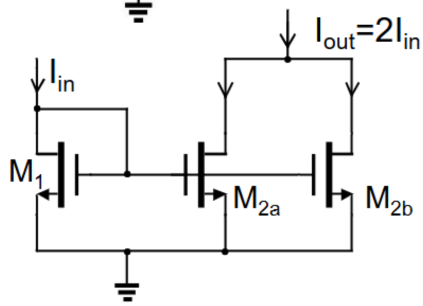


(a) $W_2=W_1$, $L_2=0.5L_1$

λ mismatch $\rightarrow$ $\dfrac{I_{out}}{I_{in}} = \dfrac{(2W/L_1)(1+\lambda_2 V_{out})}{(W/L_1)(1+\lambda_1 V_{in})} \neq 2$

(b) $W_2=2W_1$, $L_2=L_1$

ΔW and ΔL mismatch $\rightarrow$ $\dfrac{I_{out}}{I_{in}} = \dfrac{(2W+\triangle W)/(L_1+\triangle L)}{(W+\triangle W)/(L_1+\triangle L)} \neq 2$



(c) Two transistors in parallel,

$W_{2a}=W_{2b}=W_1$
$L_{2a}=L_{2b}=L_1$

$\dfrac{I_{out}}{I_{in}} = \dfrac{2(W+\triangle W)/(L_1+\triangle L)}{(W+\triangle W)/(L_1+\triangle L)} = 2$

- the error due to sizing affects MOSFET of difference size more
  - so its more prevalent in when current ratio is not the same
- We cannot change sizing ratio with different L, because that will change λ giving us more error
- Changing W only also gives error due to fabrication error
- But if we assume that the variation is the same for all transistors. Putting mosfet in parallel can eliminate that error
  - so for current mirror, we should use multiplier, not fingers
  - fingers is more like the option b
- In terms of error from the 3 ways: c < b < a

Example: Simple Current Mirror

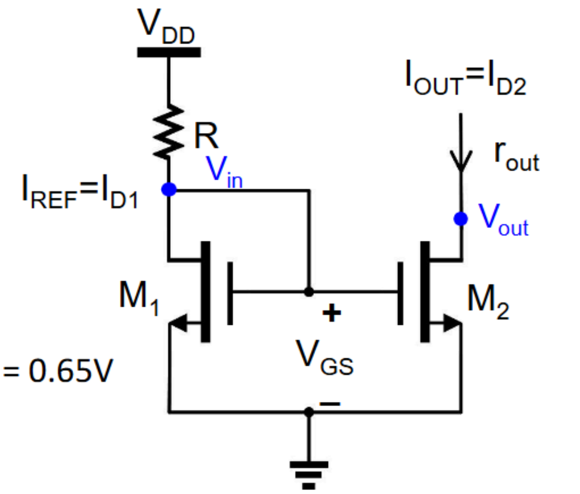
For the circuit below,

1) Find the value of R such that $I_{out} = I_{REF} = 10\mu A$.

2) Find V_{out_min}.

3) What is the condition for systematic gain error free, assuming two transistors are matched?

where V_{DD}=1.8V, V_{TH}=0.5V, $k'_n=140\times10^{-6}$ A/V², and W/L=10. Ignore channel length modulation.



1) Resistor R: $V_{OD} = \sqrt{\frac{2I_{out}}{k'_n(W/L)_2}} \approx 0.12V$

$$R = \frac{V_{DD}-V_{GS}}{I_{REF}} = \frac{V_{DD}-V_{TH}-V_{OD}}{I_{out}} = 118k\Omega$$

2) V_{out_min}: $V_{out\_min} = V_{OD} = 0.12V$

3) For gain error free: $V_{out} = V_{GS1} = V_{TH} + V_{OD} = 0.5 + 0.12 = 0.65V$

- the condition for systematic gain error free is when Vout = Vin

Cascode Current Mirrors

- From previous analysis, the current mirror is more accurate if it has matched V_{in} and V_{out}, and high output resistance.
- Now we modify the simple current mirror by adding one more transistor M_3
- Notice that the output branch is a "cascode topology". Hence, this current mirror is called a "cascode current mirror".
- Cascode topology has high output resistance

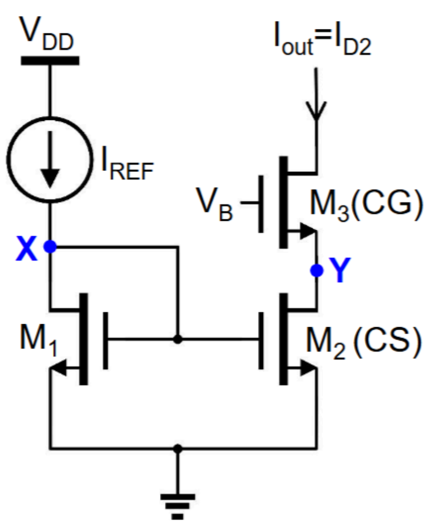
  $$r_{out} \approx (1 + g_{m3}r_{o2}) \cdot r_{o3}$$

- Note that the current mirroring is between M_1 and M_2. To eliminate the systematic gain error, we need to make V_Y=V_X.

  $$V_x = V_{GS1}$$

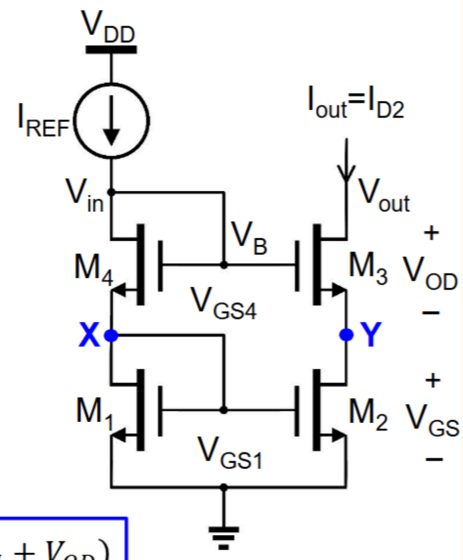
  $$V_y = V_B - V_{GS3} = V_x = V_{GS1}$$

  $$V_B = V_{GS3} + V_{GS1}$$ ← Condition to eliminate systematic gain error



- Cascode is when we put a common gate on top of M2
  - this is cascode topology
- The purpose of M3 is to boost the output resistance to even higher (because we have two r_o multiplying), and to provide us away to ensure the Vin = Vout, eliminating systematic gain error

- How to generate the required V_B in last slide?
- One simple way is to create an exact replica of M_3 in the left branch using a diode connected transistor M_4 and now V_B is ($V_{GS4}+V_{GS1}$)



<u>Analysis</u>

- Assume that all transistors are matched (same W/L, V_{TH}, etc) and ignore the body effect

  $V_{OD1} = V_{OD2} = V_{OD3} = V_{OD4} \rightarrow V_{GS1} = V_{GS2} = V_{GS3} = V_{GS4}$

  $V_B = V_{GS1} + V_{GS4} \rightarrow V_y = V_B - V_{GS3} = V_B - V_{GS4} = V_{GS1} = V_x$

- Minimum output voltage V_{out_min}:

  For M_1, M_2 and M_4, $V_{DS} = V_{GS} > V_{GS} - V_{TH} \rightarrow$ all in saturation

  For M_3 in saturation $\rightarrow V_{G3} - V_{D3} \leq V_{TH}$

  $V_B - V_{out} \leq V_{TH} \rightarrow V_{out} \geq V_B - V_{TH} = 2V_{GS} - V_{TH} = V_{TH} + 2V_{OD}$

  $\boxed{V_{out\_min} = V_{TH} + 2V_{OD}}$ $\boxed{V_{in\_min} = V_B = V_{out\_min} + V_{TH} = 2(V_{TH} + V_{OD})}$

- So to give the accurate current, we need to bias the VB very accurately, any error here will make the condition for systematic gain error free gone
- One way is to add a M4 on top of M1, and M4 must be matched to M3
- It looks like a current mirror stacked on top of another current mirror
- The price to pay is that the output swing is reduced, because the voltage headroom is decreased due to stacking of mosfet (this is always a limitation of cascode topology)
  - M1 and M4 are diode connected ⇒ always in saturation
  - M2 is guaranteed to be in saturation due to the matched M3 and M4 forcing voltage at point Y to be V_{GS1}
  - Now we need to ensure M3 is in saturation

    VDS3 ≥ VOD3
    Vout - VY ≥ VB - VY - VTH3
    Vout ≥ VB - VTH3
    Vout ≥ VGS1 + VGS4 - VTH3
    Vout ≥ VOD1 + VTH1 + VOD4 + VTH4 - VTH3
    Vout ≥ 2VOD + VTH since all mosfet is same sizing all VGS and VOD are the same
    ⇒ Vout, min = 2VOD + VTH

    When Vout = Vout, min, to ensure M3 is in saturation:
    VD3 ≥ VG3 - VTH3
    Vout, min ≥ Vin - VTH3
    2VOD + VTH ≥ Vin -VTH
    Vin ≤ 2VOD + 2VTH
    Vin, min = 2VOD + 2VTH
    ⇒ Vin, min is the max Vin can be when Vout is min, because if the Vin increase, M3 will want to conduct more, pulling the output voltage lower, making it go into linear region

- In cascode current mirror, $V_{out\_min} = V_{TH} + 2V_{OD}$
- For example, if $V_{TH} = 0.5V$ and $V_{OD} = 0.2V$, $V_{out\_min} = 0.9V$.
  The input voltage $V_{in}=2(V_{TH} + V_{OD}) = 1.4V$. Both are too high for low V_{DD}, such as 1.2V
- Thus, there is need to further reduce V_{out_min}

- The voltage headroom requirement is actually fairly high, so there is a motivation to reduce it further with a different design

First Approach We begin with the first principle and wish to ensure that V_{DS2} in Fig. 5.5(b) is both *constant* and equal to V_{DS1}. Recall from Chapter 3 that a cascode device can shield a current source, thereby reducing the voltage variations across it. As shown in Fig. 5.12(a), even though the analog circuit may allow V_P to vary substantially, V_Y remains relatively constant. But how do we ensure that $V_{DS2} = V_{DS1}$? We must generate V_b such that $V_b - V_{GS3} = V_{DS1}$ ($= V_{GS1}$), i.e., $V_b = V_{GS3} + V_{GS1}$. In other words, V_b can be established by two diode-connected devices in series [Fig. 5.12(b)], provided that $V_{GS0} + V_{GS1} = V_{GS3} + V_{GS1}$, and hence $V_{GS0} = V_{GS3}$. We now attach the V_b generator of Fig. 5.12(b) to the cascode current source as shown in Fig. 5.12(c). The result allows accurate copying of the current even in the presence of body effect (why?).

cuz M1 M2 no body effect?

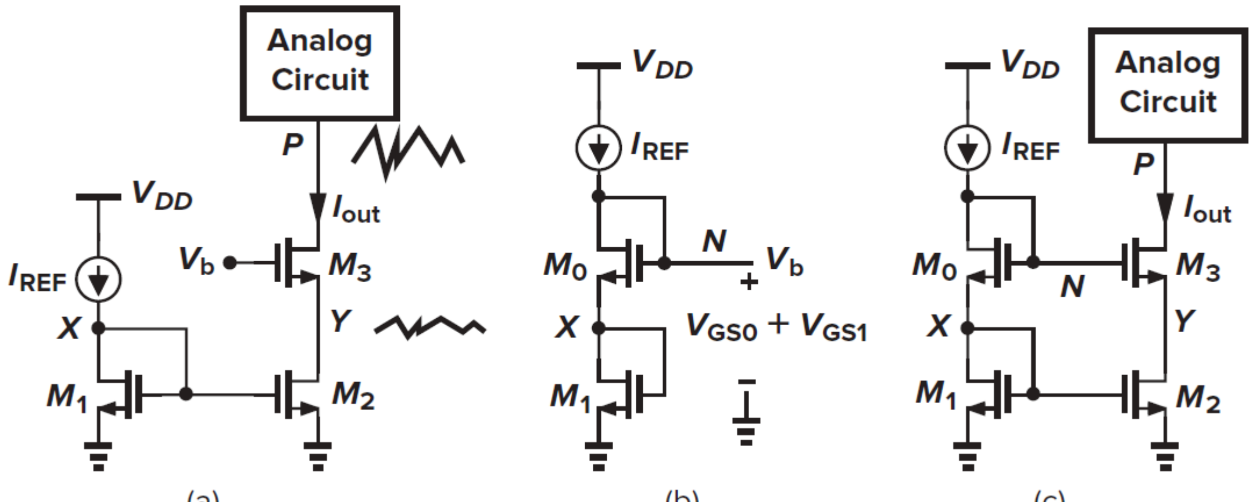


Figure 5.12 (a) Cascode current source, (b) modification of mirror circuit to generate the cascode bias voltage, and (c) cascode current mirror.

A few notes on the sizing of the transistors in Fig. 5.12(c) are warranted. As explained earlier, we typically select $L_2 = L_1$ and scale W_2 (in integer units) with respect to W_1 to obtain the desired multiple of I_{REF}. Similarly, for V_{GS3} to be equal to V_{GS0}, we choose $L_3 = L_0$ and scale W_3 with respect to W_0 by the same factor, i.e., $W_3/W_0 = W_2/W_1$. In practice, L_3 and L_0 are equal to the minimum allowable value so as to minimize their width, while L_1 and L_2 may be longer in some cases.[1]

- the act of cascading allow M3 to shield the voltage drop, so the M2 experience less voltage swing

While operating as a current source with a high output impedance and accurate value, the topology of Fig. 5.12(c) nonetheless consumes substantial voltage headroom. For simplicity, let us neglect the body effect and assume that all of the transistors are identical. Then, the minimum allowable voltage at node P is equal to

$$V_N - V_{TH} = V_{GS0} + V_{GS1} - V_{TH} \tag{5.12}$$

$$= (V_{GS0} - V_{TH}) + (V_{GS1} - V_{TH}) + V_{TH} \tag{5.13}$$

i.e., two overdrive voltages plus one threshold voltage. How does this value compare with that in Fig. 5.12(a) if V_b could be chosen more arbitrarily? As shown in Fig. 5.14(a), V_b could be so low

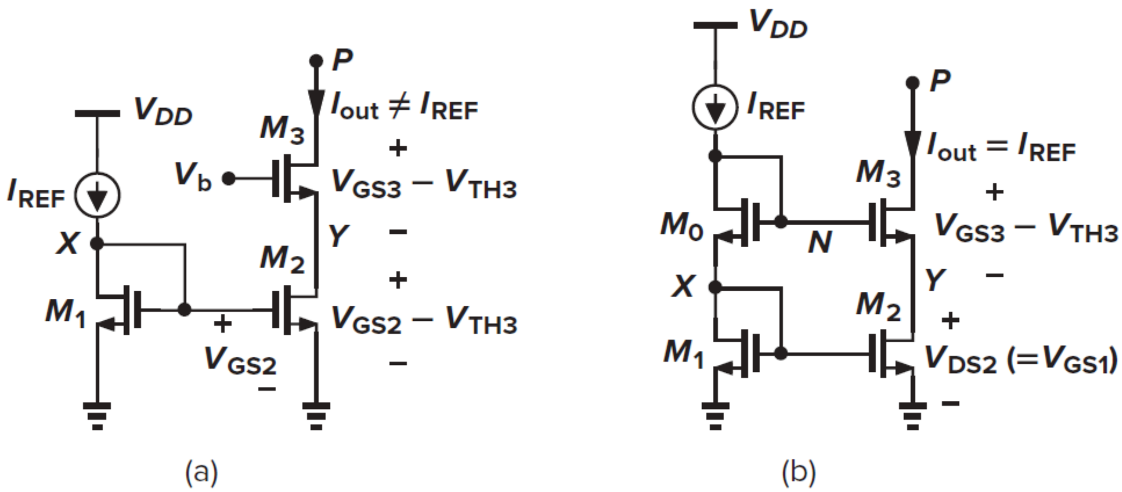
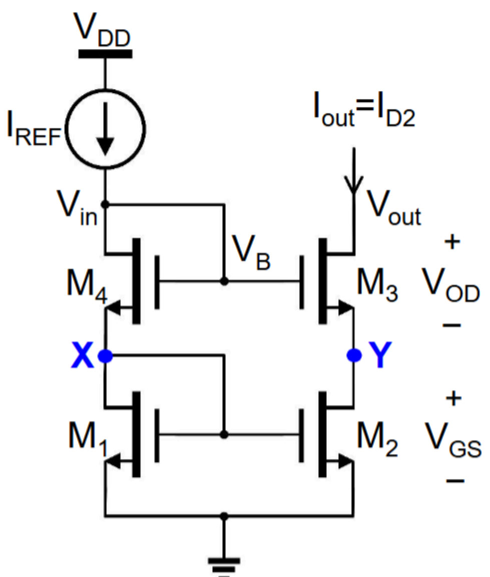


Figure 5.14 (a) Cascode current source with minimum headroom voltage; (b) headroom consumed by a cascode mirror.

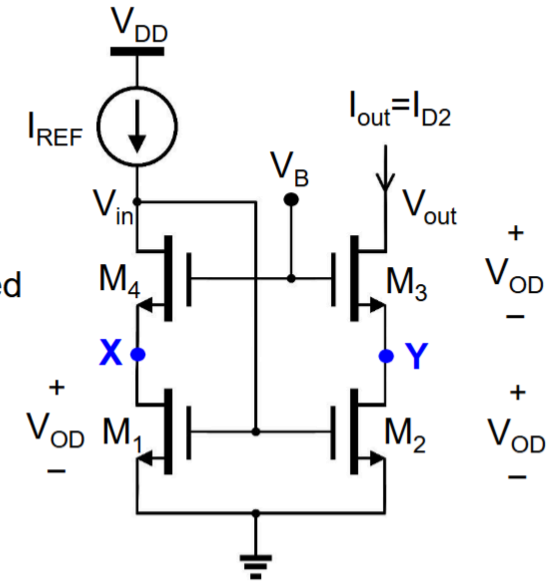
(= $V_{GS3} + V_{GS2} - V_{TH2}$) that the minimum allowable voltage at P is merely two overdrive voltages. Thus, the cascode mirror of Fig. 5.12(c) "wastes" one threshold voltage in the headroom. This is because $V_{DS2} = V_{GS2}$, whereas V_{DS2} *could* be as low as $V_{GS2} - V_{TH}$ while maintaining M_2 in saturation.

Figure 5.14 summarizes our discussion. In Fig. 5.14(a), V_b is chosen to allow the lowest possible value of V_P, but the output current does not accurately track I_{REF} because M_1 and M_2 sustain unequal drain-source voltages. In Fig. 5.14(b), a higher accuracy is achieved, but the minimum level at P is higher by one threshold voltage.

Low Voltage Cascode Current Mirror



We want to reduce V_{DS2} to V_{OD}, so that V_{out_min} is reduced to $2V_{OD}$



- the main issue wit the old design is the when making VX = VY, we force the VDS2 to be VGS1, cucking ourselves of the possible lower VDS2 of VOD

Required value of V_B?

- V_B determines V_{DS2}. To make V_{DS2}=V_{OD},

  $V_B = V_{GS} + V_{OD} = V_{TH} + 2V_{OD}$, hence $\quad V_B \neq 2V_{GS}$
- Since V_Y=V_{OD}, and V_X=V_B-V_{GS}=V_{OD} → No gain error
- The input voltage now is V_{in} = V_{GS} =V_{TH} + V_{OD}

Are all transistors in saturation?

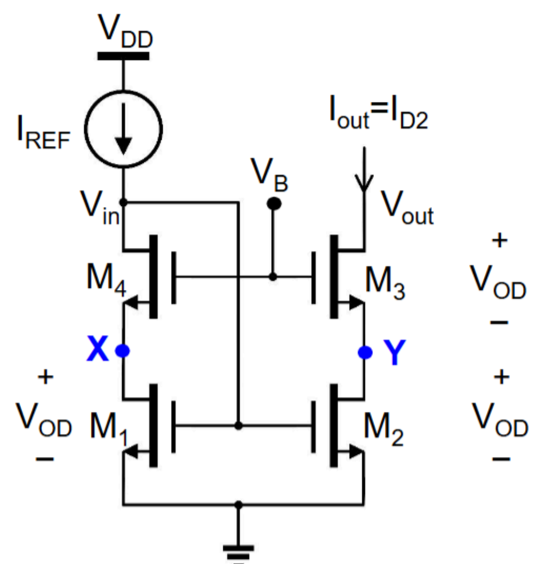
- The V_{DS} of M_1 M_2 and M_3 and all V_{OD}, thus they are in saturation.
- For M_4 to be in saturation, we have

  $V_B - V_{in} \leq V_{TH}$

  $(V_{TH} + 2V_{OD}) - (V_{TH} + V_{OD}) \leq V_{TH}$

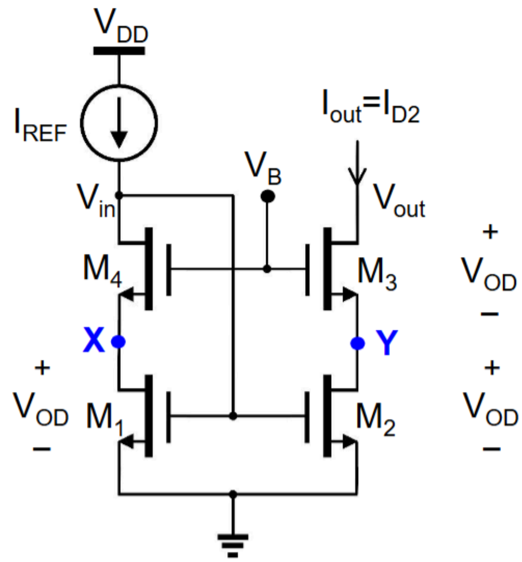
  $V_{OD} \leq V_{TH}$

If we design V_{OD4} < V_{TH}, M_4 will be in saturation



*Assume all transistors are matched and has no body effect, hence have same V_{OD}.

- Now we have a low voltage current mirror
  - Low $V_{out\_min} \rightarrow 2V_{OD}$ (as oppose to $V_{TH} + 2V_{OD}$)
- Low $V_{IN} \rightarrow V_{GS} = V_{TH} + V_{OD}$ (as oppose to $2V_{TH} + 2V_{OD}$)
  - No systematic gain error, and
  - High output resistance, $\sim g_{m3} \cdot r_{o2} \cdot r_{o3}$



- This combined the benefit of lower voltage headroom, higher output resistance, and no systematic gain error
  - acts more like an ideal current source

How to Generate VB?

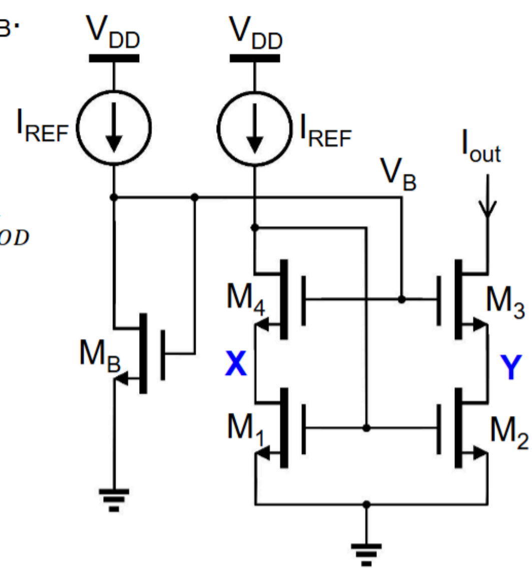
Add transistor M_B and adjust the W/L to get the required V_B.

$$I_{REF} = \frac{1}{2}k_n'\left(\frac{W}{L}\right)_B (V_B - V_{TH})^2 = \frac{1}{2}k_n'\left(\frac{W}{L}\right)_1 (V_{GS1} - V_{TH})^2$$

$$V_B - V_{TH} = (V_{TH} + 2V_{OD} - V_{TH}) = 2V_{OD} \quad \text{and} \quad V_{GS1} - V_{TH} = V_{OD}$$

$$\frac{1}{2}k_n'\left(\frac{W}{L}\right)_{MB} (2V_{OD})^2 = \frac{1}{2}k_n'\left(\frac{W}{L}\right)_1 (V_{OD})^2$$

$$\rightarrow 2\left(\frac{W}{L}\right)_{MB} = \frac{1}{2}\left(\frac{W}{L}\right)_1 \rightarrow \left(\frac{W}{L}\right)_{MB} = \frac{1}{4}\left(\frac{W}{L}\right)_1$$



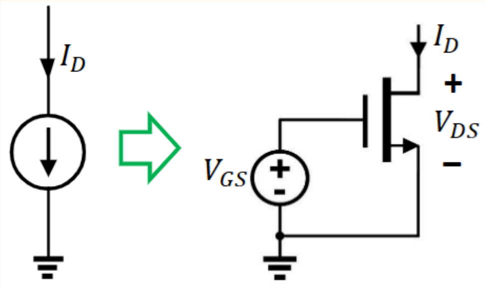
V_B can be generated by choosing channel width W_B to be ¼ W_1 with the same L.

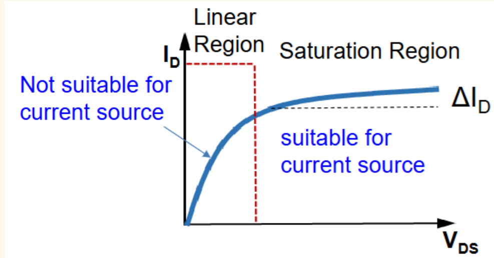
- always use current to generate voltage, because voltage is hard to generate accurately
- use another current mirror to create VB
  - we spent a lot of effort to make Vx = Vy but none to make sure this additional current mirror has the same matching voltage
  - and it is true, if we simulate, VB will not be very accurate
  - BUT this design will be more accurate than the simple current mirror that does not make Vx = Vy

Current Mirror (CM)	r_{out}	V_{omin}	V_{IN}	**Systematic gain error**	**Transistor count**
Simple CM	r_o	V_{OD}	$V_{TH}+V_{OD}$	$\lambda(V_{out}-V_{in})$	2
Cascode CM	$\sim g_m 3 r_{o2} r_{o3}$	$V_{TH}+2V_{OD}$	$2V_{TH}+2V_{OD}$	None	4
L-V Cascode CM with V_B-1	$\sim g_m 3 r_{o2} r_{o3}$	$2V_{OD}$	$V_{TH}+V_{OD}$	None	5
L-V Cascode CM with V_B-2	$\sim g_m 3 r_{o2} r_{o3}$	$2V_{OD}$	$V_{TH}+V_{OD}$	None	7

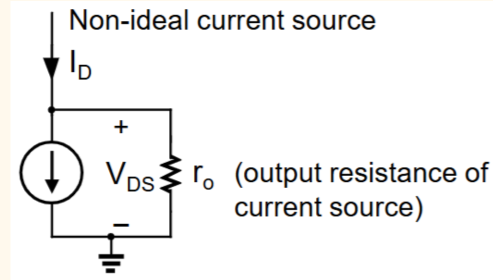
Summary

Realistic Current Source





This limits the output voltage swing



Ideal: r_o is infinite

Mismatch Error (Random Error)

Mismatch due to fabrication for devices designed to be matched

- W, L, doping concentration, mask misalignment

- A current mirror with current ratio of two is designed. Assume $V_{IN}=V_{OUT}$ to eliminate the channel length modulation effect.
- Due to mismatch between M_1 and M_2, the output current may not be exactly two times of the input current. See examples below.
- 3 ways to implement: (assumes same ΔW and ΔL for all transistors)

(a) $W_2=W_1$, $L_2=0.5L_1$

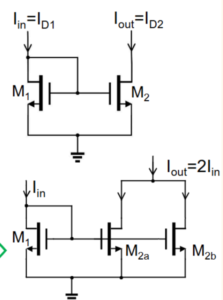
λ mismatch $\rightarrow$ $\frac{I_{out}}{I_{in}} = \frac{(2W/L_1)(1+\lambda_2 V_{out})}{(W/L_1)(1+\lambda_1 V_{in})} \neq 2$

(b) $W_2=2W_1$, $L_2=L_1$

ΔW and ΔL mismatch $\rightarrow$ $\frac{I_{out}}{I_{in}} = \frac{(2W+\Delta W)/(L_1+\Delta L)}{(W+\Delta W)/(L_1+\Delta L)} \neq 2$

(c) Two transistors in parallel, $W_{2a}=W_{2b}=W_1$, $L_{2a}=L_{2b}=L_1$

$\frac{I_{out}}{I_{in}} = \frac{2(W+\Delta W)/(L_1+\Delta L)}{(W+\Delta W)/(L_1+\Delta L)} = 2$



Assuming that error for dimension is fixed for all devices, using two transistor in parallel (multiplier not fingers), can eliminate the random error

- In terms of error from the 3 ways: c < b < a

Gain Error (Systematic Error)

Error due to circuit design (assuming stuff that is not true, like ignoring λ)

Systematic gain error definition: $E_{rr} = \frac{I_{OUT}-I_{OUT,ideal}}{I_{OUT,ideal}}$

Assume $\lambda_1 = \lambda_2$

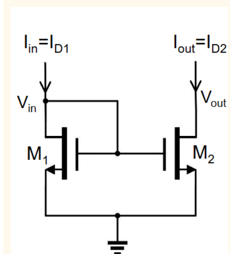
$$\frac{I_{out}}{I_{in}} = \frac{I_{D2}}{I_{D1}} = \frac{\left(\frac{W}{L}\right)_2 (1+\lambda V_{out})}{\left(\frac{W}{L}\right)_1 (1+\lambda V_{in})} \qquad I_{out} = \frac{\left(\frac{W}{L}\right)_2 (1+\lambda V_{out})}{\left(\frac{W}{L}\right)_1 (1+\lambda V_{in})} I_{in}$$

Gain error:

$I_{out,ieeal} = \frac{(W/L)_2}{(W/L)_1} I_{in}$ and assume $\frac{(W/L)_2}{(W/L)_1} = 1$

$$E_{rr} = \frac{I_{out}-I_{out,ideal}}{I_{out,ideal}} = \frac{(1+\lambda V_{out})}{(1+\lambda V_{in})} - 1 = \frac{\lambda}{1+\lambda V_{in}}(V_{out}-V_{in})$$

If $\lambda V_{in} \ll 1$ $\quad E_{rr} \approx \lambda(V_{out}-V_{in})$

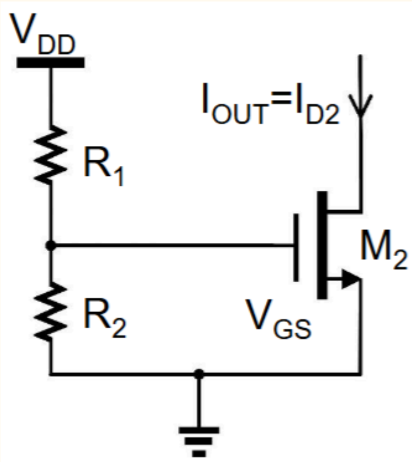


this is the error due to λ, or due to the difference of the two voltages

- reduction in error with big L and small λ
- elimination of error with Vout = Vin

Designs

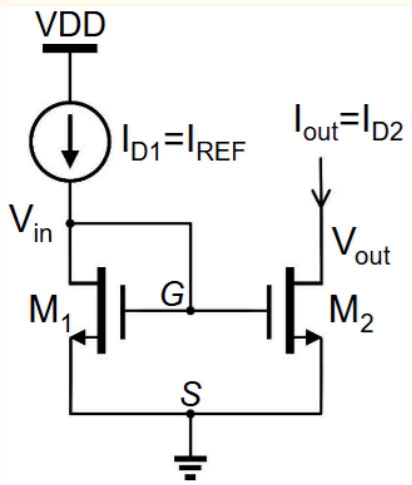
Resistor Divider Biasing



$$I_{D2} = \frac{1}{2}\left(\frac{W}{L}\right)_2 \mu C_{ox}(V_{GS} - V_{TH})^2\ (1 + \lambda V_{DS2})$$
$$= \frac{1}{2}\left(\frac{W}{L}\right)_2 \mu C_{ox}\left(\frac{V_{DD}\cdot R_2}{R_1+R_2} - V_{TH}\right)^2\ (1 + \lambda V_{DS2})$$

- Terrible design, sensitive to VDD, R1, R2, VTH which all changes with PVT (process, voltage and temperature)
- Fix this with current mirror

Simple Current Mirror



Equate square law through V_{GS}

$$I_{REF} = I_{D1} = \frac{1}{2}k_n'\left(\frac{W}{L}\right)_1 (V_{GS} - V_{TH})^2(1 + \lambda V_{DS1})$$
$$I_{out} = I_{D2} = \frac{1}{2}k_n'\left(\frac{W}{L}\right)_2 (V_{GS} - V_{TH})^2(1 + \lambda V_{out})$$

Assuming V_{TH} is the same (in reality it is not for different sizing)

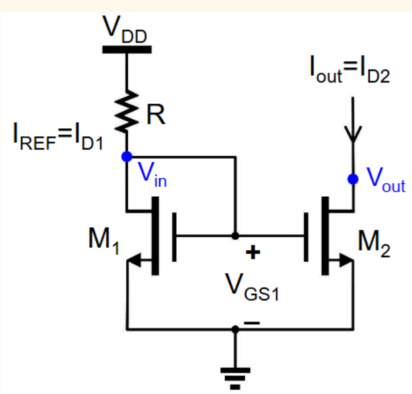
$$\frac{I_{out}}{I_{REF}} = \frac{I_{D2}}{I_{D1}} = \frac{\left(\frac{W}{L}\right)_2 (1 + \lambda V_{out})}{\left(\frac{W}{L}\right)_1 (1 + \lambda V_{GS1})}$$

Assuming Vin = Vout (in reality it is not due to output voltage swing), λ can be made same by choosing same L

$$I_{out} = \frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1} I_{REF} \qquad (V_{in} = V_{out})$$

This causes systematic gain error

Resistor for Current Mirror



For diode-connected transistor M_1: $V_{DS1} = V_{GS1} > V_{GS1} - V_{TH1}$

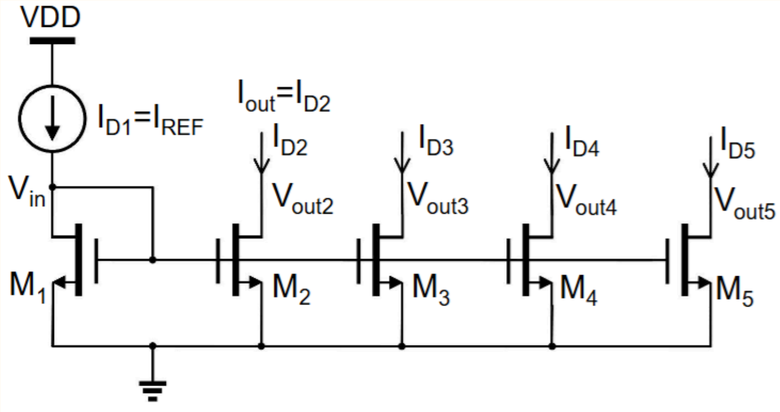
Hence, M_1 is always in saturation region

$$I_{D1} = \frac{1}{2}\mu C_{ox}\frac{W}{L}(V_{GS1} - V_{TH1})^2\ (1 + \lambda V_{DS1}) = \frac{V_{DD} - V_{GS1}}{R}$$

Use Ohm's and Square Law to solve

Has all the same assumption and error as simple current mirror above

Cascaded Simple Current Mirror



$$I_{Dx} = \frac{\left(\frac{W}{L}\right)_x}{\left(\frac{W}{L}\right)_1} I_{REF}$$

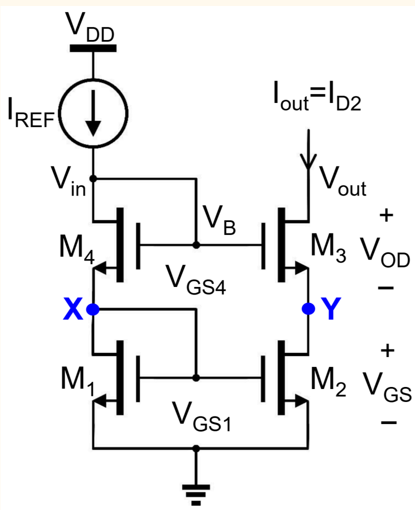
$$\alpha_x = \frac{\left(\frac{W}{L}\right)_x}{\left(\frac{W}{L}\right)_1}$$

x=2, 3, 4 ,5…

Assume $V_{outx} = V_{GS1}$

Assuming V_{TH} is the same and Vin = Vout, has systematic gain error

<u>Cascode Current Mirror</u>



Requires high V_{DD} due to $V_{out,min}$

Cascode topology has high output resistance

$$r_{out} \approx (1 + g_{m3} r_{o2}) \cdot r_{o3}$$

Note that the current mirroring is between M_1 and M_2. To eliminate the systematic gain error, we need to make V_Y=V_X.

$$V_x = V_{GS1}$$

$$V_y = V_B - V_{GS3} = V_x = V_{GS1}$$

$$V_B = V_{GS3} + V_{GS1}$$ ← Condition to eliminate systematic gain error

Assume that all transistors are matched (same W/L, V_{TH}, etc) and ignore the body effect

$$V_{OD1} = V_{OD2} = V_{OD3} = V_{OD4} \rightarrow V_{GS1} = V_{GS2} = V_{GS3} = V_{GS4}$$

$$V_B = V_{GS1} + V_{GS4} \rightarrow V_y = V_B - V_{GS3} = V_B - V_{GS4} = V_{GS1} = V_x$$

Minimum output voltage V_{out_min}:

For M_1, M_2 and M_4, $V_{DS} = V_{GS} > V_{GS} - V_{TH}$ → all in saturation

For M_3 in saturation → $V_{G3} - V_{D3} \le V_{TH}$

$$V_B - V_{out} \le V_{TH} \rightarrow V_{out} \ge V_B - V_{TH} = 2V_{GS} - V_{TH} = V_{TH} + 2V_{OD}$$

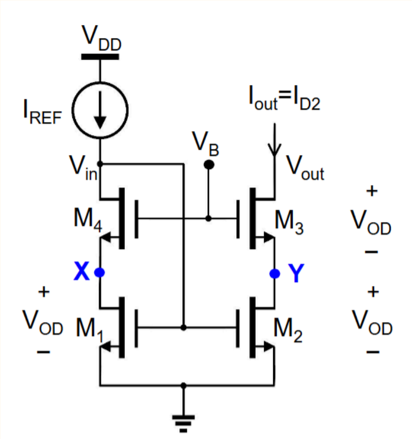
$$\boxed{V_{out\_min} = V_{TH} + 2V_{OD}} \qquad \boxed{V_{in\_min} = V_B = V_{out\_min} + V_{TH} = 2(V_{TH} + V_{OD})}$$

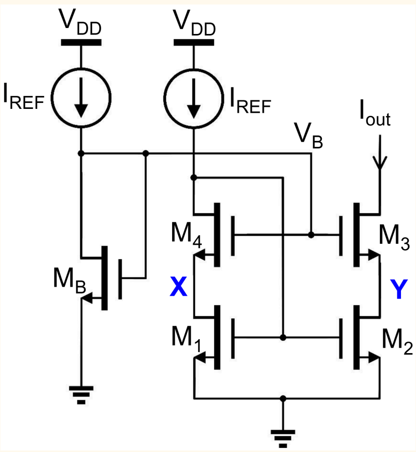
⇒ $V_{in,\,max} = 2\,(V_{OD} + V_{TH})$

⇒ Vin, min is the max Vin can be when Vout is min, because if the Vin increase, M3 will want to conduct more, pulling the output voltage lower, making it go into linear region

<u>Low Voltage Cascode CM</u>

Assume all transistors are matched and have no body effect, have same VOD





Required value of V_B?

- V_B determines V_{DS2}. To make $V_{DS2}=V_{OD}$,
  $V_B = V_{GS} + V_{OD} = V_{TH} + 2V_{OD}$, hence $V_B \neq 2V_{GS}$
- Since $V_Y=V_{OD}$, and $V_X=V_B-V_{GS}=V_{OD}$ → No gain error
- The input voltage now is $V_{in} = V_{GS} = V_{TH} + V_{OD}$

Are all transistors in saturation?

- The V_{DS} of M_1 M_2 and M_3 and all V_{OD}, thus they are in saturation.
- For M_4 to be in saturation, we have
  $V_B - V_{in} \leq V_{TH}$
  $(V_{TH} + 2V_{OD}) - (V_{TH} + V_{OD}) \leq V_{TH}$
  $V_{OD} \leq V_{TH}$

If we design $V_{OD4} < V_{TH}$, M_4 will be in saturation

$$I_{REF} = \frac{1}{2}k'_n\left(\frac{W}{L}\right)_B (V_B - V_{TH})^2 = \frac{1}{2}k'_n\left(\frac{W}{L}\right)_1 (V_{GS1} - V_{TH})^2$$

$$V_B - V_{TH} = (V_{TH} + 2V_{OD} - V_{TH}) = 2V_{OD} \quad \text{and} \quad V_{GS1} - V_{TH} = V_{OD}$$

$$\frac{1}{2}k'_n\left(\frac{W}{L}\right)_{MB} (2V_{OD})^2 = \frac{1}{2}k'_n\left(\frac{W}{L}\right)_1 (V_{OD})^2$$

$$\rightarrow 2\left(\frac{W}{L}\right)_{MB} = \frac{1}{2}\left(\frac{W}{L}\right)_1 \rightarrow \left(\frac{W}{L}\right)_{MB} = \frac{1}{4}\left(\frac{W}{L}\right)_1$$

- we spent a lot of effort to make Vx = Vy but none to make sure this additional current mirror has the same matching voltage
- and it is true, if we simulate, VB will not be very accurate
- BUT this design will be more accurate than the simple current mirror that does not make Vx = Vy

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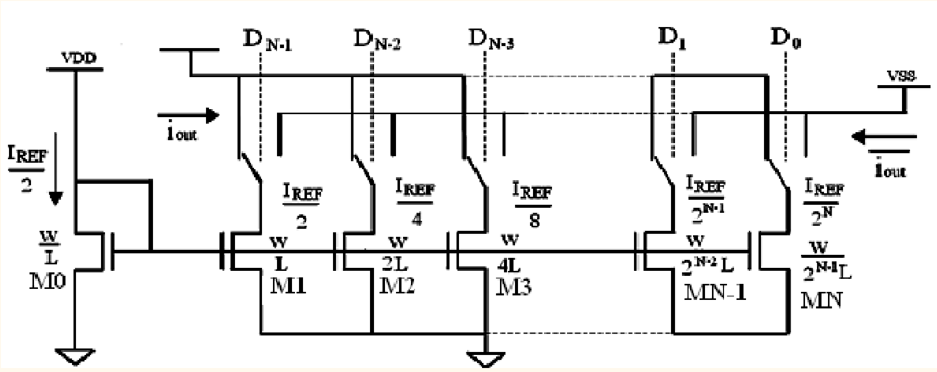
L-V Cascode CM with V_B-2	$\sim g_{m3} r_{o2} r_{o3}$	$2V_{OD}$	$V_{TH}+V_{OD}$	None	7

its g_{m3} referring to the cascode CG mosfet on top of the mirror

- the ideal with all these stacking is the same
  - Make at least one current mirror pair with the voltage constrained, this will force the current to be the same in both branches

- One way is to make the important voltage shielded by another mosfet, the other mosfet will do the voltage drop required to maintain current across load, so its voltage buffer to do so is VOD
  - But need to bias this shield mosfet properly, otherwise nothing is stopping the important voltage from wondering
    - Since all the mosfet size is the same, all the VOD and VGS the same, can use other mosfet around the circuit to constraint this voltage to be the same, Vx = Vy
- The different ways to fix this voltage is to make sure the voltage is as low as possible, so that the load voltage range is bigger
  - the target is to drop the minimal amount of VOD

Why put so much effort into getting a current mirror with 0 gain error, when we can always simulate and tune the circuit to adapt to the slightly off current?

- this is ok for circuit that does not require very precise magnitude of current
- circuit with very important magnitude of current is for example current DAC (might use triple cascode)
  - 

For circuits like differential pair the cascode CM is mainly used for its high output resistance, this gives big CMRR, as Acm is ~inversely proportional to the tail current source output resistance

Also in actual IC design, the first current reference made using band gap reference circuit, the resistor is for show only